SUBMISSION AND FORMATTING INSTRUCTIONS FOR MLSYS 2025

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ABSTRACT

This document provides a basic paper template and submission guidelines. Abstracts must be a single paragraph, ideally between 4–6 sentences long. Gross violations will trigger corrections at the camera-ready phase.

1 INTRODUCTION

People spend much of their working day in front of a screen, and much of what passes over it they will need again: the price cap they set while shopping, the command they ran right after checking their IP address, the album page they opened to copy a track list. Human memory is a poor index for this stream. Most such details are seen once, briefly, and are gone by the time they are needed [cite: human memory / re-finding studies], and recovering them by hand means guessing the right application, tab, or search query (Hu & Lee, 2020). *Episodic recall* over one's own laptop activity, i.e. retrieving the moment on screen that answers a natural-language question about the past, is therefore valuable to people, and increasingly to computer-use agents that act on a person's behalf and must recall what that person did before (Yang et al., 2026).

Screen-recall systems capture a screenshot every few seconds, or whenever the screen changes, and index it. Existing systems rely heavily on OCR text. Open-source tools such as Screenpipe, OpenRecall, and Rem index only the text extracted by OCR or the accessibility tree, as did the now-discontinued Rewind [cite: Screenpipe, OpenRecall, Rem, Rewind]. Microsoft Recall also indexes screenshots, but it is closed source and requires a dedicated 40-TOPS NPU [cite: Microsoft Recall]. Research systems that go beyond text send every screenshot to a cloud VLM for summarization (Wang & Chen, 2025), so the screen history leaves the device. A screen history holds a person's messages, documents, and account details, so it should be indexed on the device, and there OCR is attractive because text is cheap to extract, store, and embed. But OCR discards everything on the screen that is not text, such as layout, icons, images, and the visual state of a form or page, and its errors on small or stylized text propagate into the index.

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Preliminary work. Under review by the Machine Learning and Systems (MLSys) Conference. Do not distribute.

Visual document retrieval has shown that embedding page images directly with a vision-language model (VLM) can outperform OCR-then-embed pipelines on static document collections (Ma et al., 2024; Faysse et al., 2025). We find that the same holds on real screen streams. Holding the embedding model fixed (Qwen3-VL-Embedding-2B [cite: Qwen3-VL-Embedding]) and indexing either each screenshot or its OCR text, on 63,538 screenshots of one person's real computer use with 5,507 time-split test queries, screenshots reach 65.6 R@1 zero-shot versus 59.5 for OCR text, and 68.6/85.8 versus 66.1/79.4 R@1/R@10 after identical LoRA fine-tuning. Screenshots, however, are slow to encode: every frame is a full VLM forward pass over hundreds of visual tokens. On a laptop (Apple M2 Pro), embedding one screenshot takes 2.86 s on the GPU versus 1.27 s for the OCR route, and 6.61 s versus 2.44 s on the CPU. Because a recall system captures 720 frames per screen-hour and must index every one of them for as long as it runs, this gap is a permanent tax: at 2.86 s per frame, indexing alone occupies the laptop GPU for 34 minutes of every hour of screen time.

Our key observation is that most of this computation is redundant, in two ways. *Spatial redundancy*: much of a screen is blank or flat (backgrounds, margins, whitespace). *Temporal redundancy*: consecutive frames, taken 5 s apart, are often identical, and when they are not, only small regions change (typing, a moving cursor). We remove each with a matching technique. *Token pruning* removes spatial redundancy within a frame: following PixelPrune [cite: PixelPrune], we drop every 32×32 block whose pixels are predictable from its left and upper neighbors (maximum pixel difference $\leq 8/255$) before it enters the ViT. *K/V reuse* removes temporal redundancy across frames: a block counts as changed if any of its pixels differs by more than $8/255$ from when it was last computed; the ViT caches every layer's keys and values and recomputes only the changed patches, and the LLM reuses the cached keys and values of unchanged visual tokens. Both techniques are training-free and apply to the same embedding model at the original screen resolution.

Together, the two techniques cut compute from 6.17 to 1.93 TFLOP per frame (0.31×) and GPU latency from 2.86 to 0.98 s (0.34×), which makes the screenshot route faster than the OCR route on the same laptop (0.98 vs. 1.27 s on the GPU, 1.50 vs. 2.44 s on the CPU). The screenshot route still spends 1.6× the FLOPs of the OCR route; OCR is cheap in FLOPs but slow in practice, because text extraction runs on the CPU as many small, irregular operations. The accuracy cost is small. With LoRA, the techniques cost at most 0.5 R@1 and 1.9 R@10, and the screenshot index still leads OCR text by +2.0 R@1 and +4.5 R@10, with 95% confidence intervals that exclude zero. Zero-shot, token pruning costs 2.4 R@1, a loss that fine-tuning absorbs.

Cheaper encoding changes what a recall system can afford. A lower cost per frame makes larger pools, i.e. longer histories and denser capture, affordable. This matters because a real history keeps growing, and retrieval gets harder as it does: in a pilot study with 500 fixed queries, screenshot R@1 fell from 78.8 to 53.2 as the pool grew from 5K to 80K frames, while still leading OCR text at every pool size [TODO: verify with larger-pool experiments in the main setting]. And because the index returns screens rather than extracted text, it plugs directly into agentic workflows in which an agent must recall a value from the user's past to finish a task, such as re-applying the filters the user chose last month [TODO: agentic results].

In summary, we make the following contributions:

- We show on real human screen streams that, with the embedding model and training recipe held fixed, indexing screenshots retrieves significantly better than indexing their OCR text, but costs 2.3× (GPU) to 2.7× (CPU) more time per frame on a laptop.
- We identify spatial and temporal redundancy in screen streams and remove them without training, with token pruning within frames and K/V reuse across frames in both the ViT and the LLM. This cuts compute to 0.31× and latency to 0.34× of the full screenshot, making the screenshot route faster than OCR text on a laptop while it keeps a +2.0 R@1 / +4.5 R@10 lead.
- We build a time-split episodic-recall benchmark from human-annotated screen recordings of two people (PSAI-A: 117 days, 91.8 hours; PSAI-B: 57 days, 17.2 hours) and from computer-use rollouts (MyPCBench (Jang et al., 2026)), with single-hop, multi-hop, and agentic (recall-then-act) queries, where every query is checked against the person's entire history so that all matching recordings count as gold [TODO: final query counts after the updated test sets].
- We show that the cheaper index scales to larger pools [TODO: larger-pool results] and benefits agentic tasks [TODO: agentic results].

2 Electronic Submission

Submission to MLSys 2025 will be entirely electronic, via a web site (not email). Information about the submission process and LaTeX templates are available on the conference web site at:

https://www.mlsys.org/

The guidelines below will be enforced for initial submissions and camera-ready copies. Here is a brief summary:

- Submissions must be in PDF.
- The maximum paper length is **10 pages** excluding references and appendices. (pages 11 and on must contain only references).
- **Do not include author information or acknowledgements** in your initial submission.
- Your paper should be in **10 point Times font**.
- Make sure your PDF file only uses Type-1 fonts.
- Place figure captions *under* the figure (and omit titles from inside the graphic file itself). Place table captions *over* the table.
- References must include page numbers whenever possible and be as complete as possible. Place multiple citations in chronological order.
- Do not alter the style template; in particular, do not compress the paper format by reducing the vertical spaces.
- Keep your abstract brief and self-contained, one paragraph and roughly 4–6 sentences. Gross violations will require correction at the camera-ready phase. The title should have content words capitalized.
- Submissions that significantly diverge from the specified format will be rejected without review.

2.1 Submitting Papers

Paper Deadline: The deadline for paper submission that is advertised on the conference website is strict. If your full, anonymized, submission does not reach us on time, it will not be considered for publication. There is no separate abstract submission.

Anonymous Submission: MLSys uses double-blind review: no identifying author information may appear on the title page or in the paper itself. Section 3.3 gives further details.

Simultaneous Submission: MLSys will not accept any paper which, at the time of submission, is under review for

another conference or has already been published. This policy also applies to papers that overlap substantially in technical content with conference papers under review or previously published. MLSys submissions must not be submitted to other conferences during MLSys's review period. Authors may submit to MLSys substantially different versions of journal papers that are currently under review by the journal, but not yet accepted at the time of submission. Informal publications, such as technical reports or papers in workshop proceedings which do not appear in print, do not fall under these restrictions.

Authors must provide their manuscripts in **PDF** format. Furthermore, please make sure that files contain only embedded Type-1 fonts (e.g., using the program `pdffonts` in linux or using File/DocumentProperties/Fonts in Acrobat). Other fonts (like Type-3) might come from graphics files imported into the document.

Authors using **Word** must convert their document to PDF. Most of the latest versions of Word have the facility to do this automatically. Submissions will not be accepted in Word format or any format other than PDF.

Those who use **LaTeX** should avoid including Type-3 fonts. Those using `latex` and `dvips` may need the following two commands:

```
dvips -Ppdf -tletter -G0 -o paper.ps paper.dvi
ps2pdf paper.ps
```

It is a zero following the "-G", which tells dvips to use the config.pdf file. Newer TeX distributions don't always need this option.

Using `pdflatex` rather than `latex`, often gives better results. This program avoids the Type-3 font problem, and supports more advanced features in the `microtype` package.

Graphics files should be a reasonable size, and included from an appropriate format. Use vector formats (.eps/.pdf) for plots, lossless bitmap formats (.png) for raster graphics with sharp lines, and jpeg for photo-like images.

The style file uses the `hyperref` package to make clickable links in documents. If this causes problems for you, add `nohyperref` as one of the options to the `mlsys2025` usepackage statement.

2.2 Submitting Final Camera-Ready Copy

The final versions of papers accepted for publication should follow the same format and naming convention as initial submissions, except that author information (names and affiliations) should be given. See Section 3.3.2 for formatting instructions.

The footnote, "Preliminary work. Under review by the MLSys. Do not distribute." must be modified to "*Proceedings of the 8th MLSys Conference*, Santa Clara, CA, USA, 2025. Copyright 2025 by the author(s)."

For those using the **LaTeX** style file, this change (and others) is handled automatically by simply changing `\usepackage{mlsys2025}` to

```
\usepackage[accepted]{mlsys2025}
```

Authors using **Word** must edit the footnote on the first page of the document themselves.

Camera-ready copies should have the title of the paper as running head on each page except the first one. The running title consists of a single line centered above a horizontal rule which is 1 point thick. The running head should be centered, bold and in 9 point type. The rule should be 10 points above the main text. For those using the **LaTeX** style file, the original title is automatically set as running head using the `fancyhdr` package which is included in the MLSys 2025 style file package. In case that the original title exceeds the size restrictions, a shorter form can be supplied by using

```
\mlsystitlerunning{...}
```

just before `\begin{document}`. Authors using **Word** must edit the header of the document themselves.

3 Format of the Paper

All submissions must follow the specified format.

3.1 Length and Dimensions

Papers must not exceed ten (10) pages, including all figures, tables, but excluding acknowledgements, references and appendices. Acknowledgements should be limited to grants and people who contributed to the paper. Any submission that exceeds this page limit, or that diverges significantly from the specified format, will be rejected without review.

The text of the paper should be formatted in two columns, with an overall width of 6.75 inches, height of 9.0 inches, and 0.25 inches between the columns. The left margin should be 0.75 inches and the top margin 1.0 inch (2.54 cm). The right and bottom margins will depend on whether you print on US letter or A4 paper, but all final versions must be produced for US letter size.

The paper body should be set in 10 point type with a vertical spacing of 11 points. Please use Times typeface throughout the text.

3.2 Title

The paper title should be set in 14 point bold type and centered between two horizontal rules that are 1 point thick,

with 1.0 inch between the top rule and the top edge of the page. Capitalize the first letter of content words and put the rest of the title in lower case.

3.3 Author Information for Submission

MLSys uses double-blind review, so author information must not appear. If you are using LaTeX and the `mlsys2025.sty` file, use `\mlsysauthor{...}` to specify authors and `\mlsysaffiliation{...}` to specify affiliations. (Read the TeX code used to produce this document for an example usage.) The author information will not be printed unless `accepted` is passed as an argument to the style file. Submissions that include the author information will not be reviewed.

3.3.1 Self-Citations

If you are citing published papers for which you are an author, refer to yourself in the third person. In particular, do not use phrases that reveal your identity (e.g., "in previous work (Langley, 2000), we have shown ...").

Do not anonymize citations in the reference section. The only exception are manuscripts that are not yet published (e.g., under submission). If you choose to refer to such unpublished manuscripts (Author, 2018), anonymized copies have to be submitted as Supplementary Material via CMT. However, keep in mind that an MLSys paper should be self contained and should contain sufficient detail for the reviewers to evaluate the work. In particular, reviewers are not required to look at the Supplementary Material when writing their review.

3.3.2 Camera-Ready Author Information

If a paper is accepted, a final camera-ready copy must be prepared. For camera-ready papers, author information should start 0.3 inches below the bottom rule surrounding the title. The authors' names should appear in 10 point bold type, in a row, separated by white space, and centered. Author names should not be broken across lines. Unbolded superscripted numbers, starting 1, should be used to refer to affiliations.

Affiliations should be numbered in the order of appearance. A single footnote block of text should be used to list all the affiliations.

Each distinct affiliations should be listed once. If an author has multiple affiliations, multiple superscripts should be placed after the name, separated by thin spaces. If the authors would like to highlight equal contribution by multiple first authors, those authors should have an asterisk placed after their name in superscript, and the term "*Equal contribution" should be placed in the footnote block ahead of the list of affiliations. A list of corresponding authors and their emails (in the format Full Name <email@domain.com>) can follow the list of affiliations. Ideally only one or two names should be listed.

A sample file with author names is included in the mlsys2025 style file package. Turn on the `[accepted]` option to the stylefile to see the names rendered. All of the guidelines above are implemented by the LaTeX style file.

3.4 Abstract

The paper abstract should begin in the left column, 0.4 inches below the final address. The heading 'Abstract' should be centered, bold, and in 11 point type. The abstract body should use 10 point type, with a vertical spacing of 11 points, and should be indented 0.25 inches more than normal on left-hand and right-hand margins. Insert 0.4 inches of blank space after the body. Keep your abstract brief and self-contained, limiting it to one paragraph and roughly 4–6 sentences. Gross violations will require correction at the camera-ready phase.

3.5 Partitioning the Text

You should organize your paper into sections and paragraphs to help readers place a structure on the material and understand its contributions.

3.5.1 Sections and Subsections

Section headings should be numbered, flush left, and set in 11 pt bold type with the content words capitalized. Leave 0.25 inches of space before the heading and 0.15 inches after the heading.

Similarly, subsection headings should be numbered, flush left, and set in 10 pt bold type with the content words capitalized. Leave 0.2 inches of space before the heading and 0.13 inches afterward.

Finally, subsubsection headings should be numbered, flush left, and set in 10 pt small caps with the content words capitalized. Leave 0.18 inches of space before the heading and 0.1 inches after the heading.

Please use no more than three levels of headings.

3.5.2 Paragraphs and Footnotes

Within each section or subsection, you should further partition the paper into paragraphs. Do not indent the first line of a given paragraph, but insert a blank line between succeeding ones.

You can use footnotes[1] to provide readers with additional information about a topic without interrupting the flow of

[1]Footnotes should be complete sentences.

Algorithm 1 Bubble Sort

```
Input: data x_i, size m
repeat
  Initialize noChange = true.
  for i = 1 to m − 1 do
    if x_i > x_{i+1} then
      Swap x_i and x_{i+1}
      noChange = false
    end if
  end for
until noChange is true
```

the paper. Indicate footnotes with a number in the text where the point is most relevant. Place the footnote in 9 point type at the bottom of the column in which it appears. Precede the first footnote in a column with a horizontal rule of 0.8 inches.[2]

3.6 Figures

You may want to include figures in the paper to illustrate your approach and results. Such artwork should be centered, legible, and separated from the text. Lines should be dark and at least 0.5 points thick for purposes of reproduction, and text should not appear on a gray background.

Label all distinct components of each figure. If the figure takes the form of a graph, then give a name for each axis and include a legend that briefly describes each curve. Do not include a title inside the figure; instead, the caption should serve this function.

Number figures sequentially, placing the figure number and caption *after* the graphics, with at least 0.1 inches of space before the caption and 0.1 inches after it. The figure caption should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left. You may float figures to the top or bottom of a column, and you may set wide figures across both columns (use the environment `figure*` in LaTeX). Always place two-column figures at the top or bottom of the page.

3.7 Algorithms

If you are using LaTeX, please use the "algorithm" and "algorithmic" environments to format pseudocode. These require the corresponding stylefiles, algorithm.sty and algorithmic.sty, which are supplied with this package. Algorithm 1 shows an example.

[2]Multiple footnotes can appear in each column, in the same order as they appear in the text, but spread them across columns and pages if possible.

Table 1. Classification accuracies for naive Bayes and flexible Bayes on various data sets.

DATA SET	NAIVE	FLEXIBLE	BETTER?
BREAST	95.9± 0.2	96.7± 0.2	√
CLEVELAND	83.3± 0.6	80.0± 0.6	×
GLASS2	61.9± 1.4	83.8± 0.7	√
CREDIT	74.8± 0.5	78.3± 0.6	
HORSE	73.3± 0.9	69.7± 1.0	×
META	67.1± 0.6	76.5± 0.5	√
PIMA	75.1± 0.6	73.9± 0.5	
VEHICLE	44.9± 0.6	61.5± 0.4	√

3.8 Tables

You may also want to include tables that summarize material. Like figures, these should be centered, legible, and numbered consecutively. However, place the title *above* the table with at least 0.1 inches of space before the title and the same after it, as in Table 1. The table title should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left.

Tables contain textual material, whereas figures contain graphical material. Specify the contents of each row and column in the table's topmost row. Again, you may float tables to a column's top or bottom, and set wide tables across both columns. Place two-column tables at the top or bottom of the page.

3.9 Citations and References

Please use APA reference format regardless of your formatter or word processor. If you rely on the LaTeX bibliographic facility, use `natbib.sty` and `mlsys2025.bst` included in the style-file package to obtain this format.

Citations within the text should include the authors' last names and year. If the authors' names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel's pioneering work (1959). Otherwise place the entire reference in parentheses with the authors and year separated by a comma (Samuel, 1959). List multiple references separated by semicolons (Kearns, 1989; Samuel, 1959; Mitchell, 1980). Use the 'et al.' construct only for citations with three or more authors or after listing all authors to a publication in an earlier reference (Michalski et al., 1983).

Authors should cite their own work in the third person in the initial version of their paper submitted for blind review. Please refer to Section 3.3 for detailed instructions on how to cite your own papers.

Use an unnumbered first-level section heading for the references, and use a hanging indent style, with the first line of

the reference flush against the left margin and subsequent lines indented by 10 points. The references at the end of this document give examples for journal articles (Samuel, 1959), conference publications (Langley, 2000), book chapters (Newell & Rosenbloom, 1981), books (Duda et al., 2000), edited volumes (Michalski et al., 1983), technical reports (Mitchell, 1980), and dissertations (Kearns, 1989).

Alphabetize references by the surnames of the first authors, with single author entries preceding multiple author entries. Order references for the same authors by year of publication, with the earliest first. Make sure that each reference includes all relevant information (e.g., page numbers).

Please put some effort into making references complete, presentable, and consistent. If using bibtex, please protect capital letters of names and abbreviations in titles, for example, use {B}ayesian or {L}ipschitz in your .bib file.

3.10 Software and Data

We strongly encourage the publication of software and data with the camera-ready version of the paper whenever appropriate. This can be done by including a URL in the camera-ready copy. However, do not include URLs that reveal your institution or identity in your submission for review. Instead, provide an anonymous URL or upload the material as "Supplementary Material" into the CMT reviewing system. Note that reviewers are not required to look at this material when writing their review.

ACKNOWLEDGEMENTS

Do not include acknowledgements in the initial version of the paper submitted for blind review.

If a paper is accepted, the final camera-ready version can (and probably should) include acknowledgements. In this case, please place such acknowledgements in an unnumbered section at the end of the paper. Typically, this will include thanks to reviewers who gave useful comments, to colleagues who contributed to the ideas, and to funding agencies and corporate sponsors that provided financial support.

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A PLEASE ADD SUPPLEMENTAL MATERIAL AS APPENDIX HERE

Put anything that you might normally include after the references as an appendix here, *not in a separate supplementary file*. Upload your final camera-ready as a single pdf, including all appendices.